

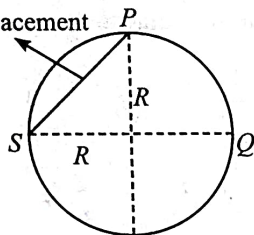
EXPLANATIONS

1. (d) Distance = $2\pi r + \pi r = 3\pi r$.

Displacement = $2r$ (straight-line distance from start to end).

2. (b) Given: $R = 2\text{ km}$

Displacement

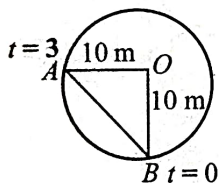


$\therefore$ Displacement = PS

$$\sqrt{R^2 + R^2} = R\sqrt{2} = 2 = \sqrt{8} \text{ km}$$

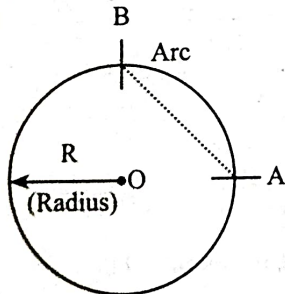
3. (d) Angle rotated by the moving body

$$\theta = \frac{3}{4} \times 2\pi = \frac{3\pi}{2}$$



$$|\text{Displacement}| = AB = \sqrt{2}R = 10\sqrt{2} \text{ m}$$

4. (a)



$$\text{Arc length} = 2\pi \left(\frac{R}{4} \right) = \frac{\pi R}{2}$$

Let instantaneous velocity be v .

Time to cover arc length

$$t = \frac{\text{Arc length}}{v} = \frac{\pi R}{2v}$$

$$AB = R\sqrt{2}$$

Average velocity,

$$<v> = \frac{AB}{t} = \frac{R\sqrt{2}(2v)}{\pi R} = \frac{2\sqrt{2}v}{\pi}$$

$$\Rightarrow \frac{v}{<v>} = \frac{\pi}{2\sqrt{2}}$$

5. (b) Distance moved, $AB = 60 \text{ m}$

$$\therefore (\theta) = \frac{\text{Arc}(AB)}{\text{radius}(R)} = \frac{3\pi}{4} = \frac{60}{R}$$

$$R = \frac{60 \times 4}{3\pi} = \frac{80}{\pi} \text{ m}$$

$$\text{Displacement} = \sqrt{R^2 + R^2 - 2R^2 \cos 135}$$

$$\{\because \cos 135 = \cos(90 + 45) = \sin 45^\circ = -0.7\}$$

$$= \sqrt{2R^2 - 2R^2(-0.7)} = \sqrt{3.4R^2}$$

$$= \sqrt{3.4 \left(\frac{80}{\pi} \right)^2} \approx 47 \text{ m}$$

6. (b) $V = V_0 + gt + Ft^2$

$$\frac{dx}{dt} = V_0 + gt + Ft^2$$

$$\Rightarrow \int_{x=0}^x dx = \int_{t=0}^{t=1} (V_0 + gt + Ft^2) dt$$

$$x = \left[V_0 t + \frac{gt^2}{2} + \frac{Ft^3}{3} \right]_{t=0}^{t=1}$$

$$\Rightarrow x = V_0 + \frac{g}{2} + \frac{F}{3}$$

7. [10] Average velocity $v_{\text{avg}} = \frac{x_1 + x_2}{t_1 + t_2}$,

$$\text{where } x_1 = x, x_2 = \frac{3}{2}x, t_1 = \frac{x}{v_1}, t_2 = \frac{\frac{3}{2}x}{v_2}$$

$$\text{Given } v_1 = 5 \text{ m/s}, v_{\text{avg}} = \frac{50}{7} \text{ m/s}$$

$$\text{Total distance} = x + \frac{3}{2}x = \frac{5}{2}x, \text{ total time}$$

$$= \frac{x}{5} + \frac{\frac{3}{2}x}{v_2}$$

$$= \frac{\frac{5}{2}x}{\frac{x}{5} + \frac{\frac{3}{2}x}{v_2}} = \frac{50}{7}$$

$$\text{Simplify: } \frac{5}{2} / \left(\frac{1}{5} + \frac{3}{2v_2} \right) = \frac{50}{7}$$

$$\text{Cross-multiply: } 5 \times 7 = 50 \left(\frac{1}{5} + \frac{3}{2v_2} \right)$$

$$35 = 10 + \frac{150}{2v_2}, 25 = \frac{150}{2v_2}$$

$$v_2 = \frac{150}{50} = 10 \text{ m/s}$$

8. (b) $x = t^3 - 6t^2 + 20t + 15$

The velocity of the body

$$\frac{dx}{dt} = v = 3t^2 - 12t + 20$$

The acceleration of the body

$$\frac{dv}{dt} = a = 6t - 12$$

When $a = 0$

$$6t - 12 = 0; t = 2 \text{ sec}$$

At $t = 2$ sec
The velocity of the body when its acceleration becomes zero
 $v = 3(2)^2 - 12(2) + 20$
 $v = 8$ m/s

9. (c) $Q = \frac{a^2 b^2}{c^2}$

By using the percentage error formula.

$$\frac{\Delta Q}{Q} = 4 \frac{\Delta a}{a} + \frac{\Delta b}{b} + 2 \frac{\Delta c}{c}$$

$$\frac{\Delta Q}{Q} = 100 = 4 \left(\frac{\Delta a}{a} \times 100 \right) +$$

$$3 \left(\frac{\Delta b}{b} \times 100 \right) + 2 \left(\frac{\Delta c}{c} \times 100 \right)$$

$$\% \text{ error in } Q = 4 \times 3\% + 3 \times 4\% + 2 \times 5\%$$

$$= 12\% + 12\% + 10\% = 34\%$$

10. [52] Given: $x = -3t^2 + 18t^2 + 16t$

$$v = -9t + 36 + 16$$

$$a = \frac{dv}{dt} = -18 + 36$$

$$a = 0 \text{ at } t = 2s$$

$$v = -9(2) + 36 + 16 = 52 \text{ m/s}$$

11. (b) Average speed

$$= \frac{\text{total distance travelled}}{\text{total time taken}}$$

$$= \frac{80 \times t_1 + 80 \times 3t_2}{4t_1} = 70 \text{ km/hr.}$$

12. [3] $x^2 = 1 + t$

Differentiating

$$2x \frac{dx}{dt} = 2t$$

$$xv = t$$

Again, differentiating

$$x \frac{dv}{dt} + v \frac{dx}{dt} = 1$$

$$x \cdot a + v^2 = 1$$

$$a = \frac{1-v^2}{x} = \frac{1-t^2/t^2}{x} = \frac{1-t^2}{x}$$

$$a = \frac{1}{x^3} = x^{-3}$$

13. (d) Given,

The particle is covering half of the distance with 6m/s

Other half distance in two equal time intervals with speed of 9m/s and 15 m/s

$$\frac{S}{t_1} = \frac{S}{t_2} = \frac{S}{t_3}$$

$$BD \Rightarrow S = 9t_1 + 15t_2 = 24t_1$$

$$AB \Rightarrow S = 6t_1 = 24t_1 \Rightarrow t_1 = 4t_2$$

$$\langle \text{speed} \rangle = \frac{\text{dist.}}{\text{time}} = \frac{48S}{2t_1 + t_2}$$

$$= \frac{48S}{2t_1 + 4t_2} = 8 \text{ m/s}$$

14. (d) Average Speed

$$= \frac{\text{Total distance of the car}}{\text{Total time}}$$

$$= \frac{x + x}{\frac{x}{v_1} + \frac{x}{v_2}} = \frac{2v_1 v_2}{v_1 + v_2}$$

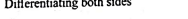
15. (a) $\therefore x = 4t^2$

Differentiating both sides

$$\frac{dx}{dt} = 8t$$

$$t = 5 \text{ sec}$$

$$v = 8 \times 5 = 40 \text{ m/s.}$$

16. [50] 

$$t_{AB} = \frac{x}{5 \text{ m/s}}$$

In motion BC

$$x = d_1 + d_2$$

where d_1 and d_2 are the distance travelled with 10 m/s and 15 m/s respectively in equal time intervals $\frac{t_1}{2}$ each

$$d_1 = \frac{10t_1}{2}, d_2 = \frac{15t_2}{2}$$

$$d_1 + d_2 = x = \frac{1}{2}(10 + 15) = \frac{25t_1}{2}$$

$$<v> = \frac{2x}{\frac{t_1}{2} + \frac{t_2}{2}} = \frac{2 \times 25}{5 + 2} = \frac{50}{7} \text{ m/s}$$

17. (d) Time for first part of motion $t_1 = \frac{4}{3}$

and time for second part of motion

$$t_2 = \frac{4}{5}$$

$$\text{Total Time } T = t_1 + t_2 = \frac{32}{15}$$

$$\therefore V_{avg} = \frac{15}{4} = 3.75 \text{ km/h}$$

18. (c) Given $AB = x$ and $BC = x$

$$\therefore AB + BC + CD = 3x$$

$$\Rightarrow 2x + CD = 3x$$

$$\therefore CD = x$$

Average speed

$$<v> = \frac{3x}{\frac{x}{v_1} + \frac{x}{v_2} + \frac{x}{v_3}} = \frac{3v_1 v_2 v_3}{v_1 v_2 + v_1 v_3 + v_2 v_3}$$

19. (d) Average velocity

$$= \frac{v \sin \left(\frac{\theta}{2} \right) + 2v \sin \left(\frac{\theta}{2} \right)}{\theta}$$

$$= \frac{2v \sin 60^\circ}{3} = 3 \sin 60^\circ = 1.5\sqrt{3} \text{ m/s}$$

20. (c) Average velocity = $\frac{\text{Total distance}}{\text{Total time}}$

$$v = \frac{2x}{\frac{x}{v_1} + \frac{1}{v_2}} = \frac{2}{\frac{1}{v_1} + \frac{1}{v_2}}$$

21. (d) Distance, $s = (2.5)^2 t^2$

$$\text{Speed } v = \frac{ds}{dt} = \frac{d}{dt} (2.5^2 t^2) = 2 \times 2.5t$$

$$v = 5t \quad \text{At } t = 5, v = 25 \text{ m/s.}$$

22. (c) $x = 5t^2 - 4t + 5$

$$\Rightarrow v = \frac{dx}{dt} = \frac{d}{dt} (5t^2 - 4t + 5) = 10t - 4$$

$$\text{At } t = 2s \text{ and } v = 16 \text{ m/s}$$

23. [18] Let the distance $AB = L$.

Then, according to the question,

$$t_1 = \frac{L}{3v_1}, t_2 = \frac{L}{3v_2}, t_3 = \frac{L}{3v_3}$$

$$V_{avg} = \frac{L}{t_1 + t_2 + t_3} = \frac{L}{\frac{L}{3v_1} + \frac{L}{3v_2} + \frac{L}{3v_3}}$$

24. (d) $x = at + bt^2 - ct^3$

$$v = \frac{dx}{dt} = a + 2bt - 3ct^2$$

$$a = \frac{dv}{dt} = 2b - 6ct = 0 \Rightarrow t = \frac{b}{3c}$$

$$v \left(\text{at } t = \frac{b}{3c} \right) = a + 2b \left(\frac{b}{3c} \right) - 3c \left(\frac{b}{3c} \right)^2$$

$$= a + \frac{2b^2}{3c} - \frac{b^2}{3c} = a + \frac{b^2}{3c}$$

25. (b) $v = b\sqrt{x}$

$$\frac{dv}{dt} = \frac{b}{2\sqrt{x}} \quad a = \frac{bv}{2\sqrt{x}}$$

$$a = \frac{b(b\sqrt{x})}{2\sqrt{x}} \Rightarrow \frac{dv}{dt} = a = \frac{b^2}{2}$$

26. (Bonus) $v^2 - u^2 = 2aS$

$$\left(\frac{2u}{3} \right)^2 = u^2 + 2(-a)(4 \times 10^{-2})$$

$$\Rightarrow \frac{4u^2}{9} = u^2 - 2a(4 \times 10^{-2})$$

$$\Rightarrow \frac{5u^2}{9} = -2a(4 \times 10^{-2}) \quad \dots(1)$$

$$\Rightarrow 0 = \left(\frac{2u}{3} \right)^2 + 2(-a)(x)$$

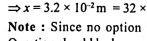
$$\Rightarrow \frac{4u^2}{9} = -2ax \quad \dots(2)$$

Dividing equation (1) by (2), we get

$$\frac{5}{4} = \frac{4 \times 10^{-2}}{x} \Rightarrow x = \frac{16}{5} \times 10^{-2}$$

$$\Rightarrow x = 3.2 \times 10^{-2} \text{ m} = 32 \times 10^{-3} \text{ m}$$

Note : Since no option is matching, Question should be bonus.

27. [45.00] 

From A $\rightarrow$ B

$$-80 = -v_1 t - \frac{1}{2} \times 10 t^2$$

$$-80 = -2v_1 - \frac{1}{2} \times 10 \times 2^2$$

$$-80 = -2v_1 - 20 \Rightarrow v_1 = 30 \text{ m/s}$$

From O to A

$$v^2 = u^2 + 2gS$$

$$30^2 = 0 + 2 \times (-10)(-S) \Rightarrow 900 = 20S$$

$$S = 45 \text{ m}$$

28. (b) Given that

$$m = 900 \text{ gm} = \frac{900}{1000} \text{ kg} = \frac{9}{10} \text{ kg}$$

$$r = 1 \text{ m}$$

$$\omega = \frac{2\pi N}{60} = \frac{2\pi(10)}{60} = \frac{\pi}{3} \text{ rad/sec}$$

The tension in the string,

$$T - mg = m\omega^2 r$$

$$T = mg + m\omega^2 r$$

$$= \frac{9}{10} \times 9.8 + \frac{9}{10} \times 1 \times \left(\frac{\pi}{3} \right)^2 = 8.82 + 0.98$$

$$= 9.80 \text{ N}$$

29. [8] Given: $V = 4\sqrt{x}$

$$a = V \frac{dv}{dx}$$

$$= 4\sqrt{x} \times 4 \times \frac{1}{2} x^{-1/2} = 8 \text{ m/s}^2$$

30. (b) $S_1 = u + \frac{a}{2}(2n-1)$

$$\Rightarrow 102.5 = u + \frac{a}{2}(2n-1) \quad \&$$

$$115 = u + \frac{a}{2}(2n+3)$$

$$\Rightarrow 12.5 = 2a \Rightarrow a = 6.25 \text{ m/s}^2$$

31. [40] Given,

$$u = 72 \text{ km/h} = 72 \times \frac{5}{18} = 20 \text{ m/s}$$

$$v = u + at$$

$$0 = 20 + a \times 4$$

$$a = -5 \text{ m/s}^2$$

$$v^2 - u^2 = 2as \Rightarrow 0^2 - 20^2 = 2(-5)s$$

$$s = 40 \text{ m}$$

32. [19] By using the formula of distance travelled in n^{th} second

$$S_n = \frac{1}{2}a(2n-1) = \frac{19a}{2}$$

Similarly

$$S_{n-1} = \frac{1}{2}a(2n-3) = \frac{17a}{2}$$

$$\frac{S_n - 1}{S_n} = \frac{17}{19} = 1 - \frac{2}{x} \Rightarrow x = 19$$

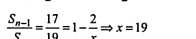
33. (a) $t_1 = \frac{u + \sqrt{u^2 + 2gh}}{g}$

$$t_2 = \frac{-u + \sqrt{u^2 + 2gh}}{g}$$

$$t = \frac{\sqrt{2gh}}{g}$$

$$t_1 t_2 = \frac{(u^2 + 2gh) - u^2}{g^2} = \frac{2gh}{g^2} = \frac{2h}{g}$$

$$\Rightarrow t = \sqrt{t_1 t_2}$$

34. (c) 

Relative velocity, $|\vec{v}_{BA}| = 40 \text{ m/s}$

Relative acceleration, $|\vec{a}_{BA}| = 4 \text{ m/s}^2$

by using the equation, $(v^2 = u^2 + 2as)$

$$0 = (40)^2 + 2(-4)(S)$$

$$S = 200 \text{ m}$$

Distance between them

$$= 300 - 200 = 100 \text{ m}$$

35. [200] $u = 20 \text{ m/s}, S_1 = 500 \text{ m}, v = 0$

Using third equation of motion

$$0 = (20)^2 - 2 \times 500 \Rightarrow a = \frac{4}{10} \text{ m/s}^2$$

After brakes are applied

$$u = 20 \text{ m/s}, S_2 = 250 \text{ m}, v = ?$$

$$v^2 = (20)^2 - 2 \times 250 = v = \sqrt{200} \text{ m/s}$$

$$\therefore x = 200$$

36. (d) Using law of motion's equation

$$v = u + at$$

$$60 = 10 + 2t \Rightarrow 2t = 50$$

$$t = 25 \text{ sec.}$$

37. (c) According to question,

$$u = 0 \quad t = 0 \quad t = t \quad t = 2t$$

$$10 = \frac{1}{2}at^2 \quad \dots(i)$$

$$10 + x = \frac{1}{2}a(2t)^2 \quad \dots(ii)$$

On subtracting (i) from (ii), we get

$$\Rightarrow x = \frac{1}{2}a[(2t)^2 - t^2] = 3 \left(\frac{1}{2}at^2 \right) = 30 \text{ m}$$

From (i) $\frac{1}{2}at^2 = 10$

Hence, option (c) is the correct answer.

38. [3] From Kinematic equation,

$$v^2 = u^2 - 2aS \Rightarrow 0 = u^2 - 2aS$$

$$S = \frac{u^2}{2a} \Rightarrow S \propto u^2$$

$$\therefore \frac{S_2}{S_1} = \left(\frac{u_2}{u_1} \right)^2 \Rightarrow S_2 = 27 \times \left(\frac{1}{3} \right)^2 = 3$$

39. [100] It is given that $\frac{dv}{dt} = 5$

So, acceleration of particle is given by

$$a = v \frac{dv}{dx}$$

$$a = 100 \text{ m/s}^2 \Rightarrow (v = 20 \text{ m/s})$$

40. (b) Let the length of train be l and its acceleration be a .

From kinematic equation,

$$\Rightarrow al = \frac{v^2 - u^2}{2}$$

Velocity when middle point crosses the post,

$$v_m = \sqrt{u^2 + 2a \frac{l}{2}} = \sqrt{u^2 + \frac{a l}{2}} = \sqrt{\frac{u^2 + v^2}{2}}$$

41. (c) $\frac{\sqrt{2l}}{\sqrt{a_2}} - \frac{\sqrt{2l}}{\sqrt{a_1}} = t$

$$S = 200 \text{ m}$$

$$\Rightarrow \frac{\sqrt{2l}}{t} = \frac{\sqrt{a_2 a_1}}{\sqrt{a_2 - a_1}}$$

68. (a) As slope of $B >$ Slope of A
 $\therefore$ Speed of $B >$ Speed of $A \Rightarrow t_B < t_A$
 69. (c) Area under the graph from $t = 0$ to $t = 20$ sec

$$A_1 = \frac{1}{2} \times 5 \times 10 + (5 \times 10) + \frac{1}{2} \times (10 + 20) \times 5 + \frac{1}{2} \times 5 \times 20 = 200 \text{ m}^2$$

Area under the graph from $t = 20$ to $t = 25$ sec

$$A_2 = \frac{1}{2} \times 5 \times (20) = 50 \text{ m}^2$$

So distance covered = $A_1 + A_2 = (200 + 50) \text{ m} = 250 \text{ m}$

Displacement = $A_1 - A_2 = (200 - 50) \text{ m} = 150 \text{ m}$

Ratio = $5/3$

70. (a) Slope of $v-t$ graph is constant but negative for part AM. Slope of $v-t$ graph is constant but positive for part MB.

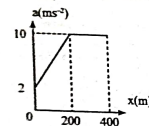
Hence, option (a) is the correct slope of the acceleration-time graph.

71. (d) $\begin{cases} 10 + \frac{x}{5}, & \text{if } 0 \leq x \leq 200 \text{ m} \\ 50, & \text{if } 200 \text{ m} \leq x \leq 400 \text{ m} \end{cases}$

$$\Rightarrow a = \frac{dv}{dt} = v \frac{dv}{dx}$$

$$= \begin{cases} \frac{1}{5} \left(10 + \frac{x}{5} \right), & \text{if } 0 \leq x \leq 200 \text{ m} \\ 0, & \text{if } 200 \text{ m} \leq x \leq 400 \text{ m} \end{cases}$$

$$\Rightarrow a = \begin{cases} \left(2 + \frac{x}{25} \right), & \text{if } 0 \leq x \leq 200 \text{ m} \\ 0, & \text{if } 200 \text{ m} \leq x \leq 400 \text{ m} \end{cases}$$



72. (b) Acceleration (a) is constant
 $\therefore v \propto t$ (Straight line graph)
 And $x \propto t^2$ (parabolic graph)

73. (a) The slope of the given v versus x graph is $m = -\frac{v_0}{x_0}$ and intercept is $c = +v_0$. Hence, varies with x as

$$v = \left(\frac{v_0}{x_0} \right) x + v_0 \quad \dots (i)$$

Using equation (i) in equation (ii), we get

$$a = v \frac{dv}{dx} = \left(\frac{-v_0}{x_0} \right) \left(\frac{-v_0}{x_0} x + v_0 \right)$$

$$\Rightarrow a = \left(\frac{v_0}{x_0} \right)^2 x - \frac{v_0^2}{x_0}$$

Thus the graph of a versus x is a straight line having a positive slope = $\left(\frac{v_0}{x_0} \right)^2$ and

$$\text{negative intercept} = -\frac{v_0^2}{x_0}$$

74. [1] As, $y = mx + c$
 $v^2 = 2x + 20$

$$v^2 = 2x + 20 \Rightarrow 2v \frac{dv}{dt} = 2v$$

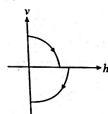
$$\Rightarrow a = 1 \text{ m/s}^2$$

75. [20]

Distance travelled = Area under the $u-t$ graph

$$\therefore \Delta S = \frac{1}{2} \times 5 \times 8 = 20$$

76. (b) At $H = h$, $v = 0$



$$\text{At } h = 0, v = \sqrt{2gh}$$

Also $a = -g$, throughout this motion

77. (a) Distance = $|A_1| + |A_2|$

$$= \frac{1}{2} \left(1 + \frac{13}{3} \right) \times 4 + \frac{1}{2} \times \frac{5}{3} \times 2 = \frac{37}{3} \text{ m}$$

78. (d) Area under $v-t$ curve

$$= \left(\frac{1}{2} \times 2 \times 2 + 2 \times 2 + 3 \times 1 \right) \text{ m} \\ = (2 + 4 + 3) = 9 \text{ m}$$

79. (c) We have, $u = 0$ and $a = \text{constant}$
 At time t ,
 $v = 0 + at \Rightarrow v = at$

$$\text{Also } x = 0(t) + \frac{1}{2} at^2 \Rightarrow x = \frac{1}{2} at^2$$

Graph (a), (b) and (d) are correct.

80. (d) velocity = $\frac{dx}{dt} = 2tC_0 + 2C_0(t-2)$

$$\text{acceleration} = \frac{d(\text{velocity})}{dt} = 2C_0 + 2C_0$$

81. (c) Given, $t = mx^2 + nx$

$$\frac{dt}{dx} = 2mx + n \Rightarrow \frac{dx}{dt} = \frac{1}{2mx + n}$$

$$v = \frac{1}{2mx + n}$$

Retardation of the motion.

$$a = v \frac{dv}{dx} = v \frac{-2m}{(2mx + n)^2} = -2mv^3$$

82. (b) $v = at + \beta t^2$

$$\frac{ds}{dt} = at + \beta t^2 \Rightarrow ds = (at + \beta t^2) dt$$

Applying integration both side.

$$\int_0^s ds = \int_0^t (at + \beta t^2) dt$$

$$s = \left[\frac{at^2}{2} + \frac{\beta t^3}{3} \right]_0^t$$

$$s = \left[2\alpha + \frac{8\beta}{3} \right] - \left[\frac{\alpha}{2} + \frac{\beta}{3} \right]$$

$$= \frac{3\alpha}{2} + \frac{7\beta}{3}$$

83. [3] We have given, the distance covered by particle varies with time t , $x^2 = at^2 + 2bt + c$

$$2xv = 2at + 2b \Rightarrow xv = at + b$$

differentiation again

$$v^2 + a = a$$

$$ac = a - \left(\frac{at+b}{x} \right)$$

$$a = \frac{a(at^2 + 2bt + c) - (at+b)^2}{x^3}$$

$$a = \frac{ac - b^2}{x^3}; a \propto x^{-3}$$

Hence, value of n is 3.

84. [3] Acceleration of $P = P = kt$, k is constant
 Acceleration of $Q = a$, a is constant
 acceleration of Q w.r.t. P

$$= a_Q - a_P = a - kt$$

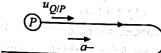
$$u_{QP} = u_Q - u_P$$

Case-I

$$u_{QP} \text{ and } a_{QP}$$

$$t = 0$$

As observed from P two cases are possible



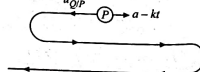
Total number of crossing = 2

Case-II

u_{QP} and a_{QP} in opposite direction

As observed from P

$$t = 0$$



Total number of crossing = 3

85. (c) Given:

$$V_A = 20 \text{ m/s}$$

$$V_B = -30 \text{ m/s}$$

Velocity of B w.r.t. A

$$V_{BA} = -50 \text{ m/s}$$

Velocity of ground w.r.t. B

$$V_{GB} = 30 \text{ m/s}$$

$$u_B = 20 \text{ m/s}$$

$$\therefore \text{time}(t) = \frac{\text{distance}(s)}{\text{speed}(n)}$$

$$\therefore t_B - t_A = 35$$

$$\frac{S_B}{u_B} - \frac{S_A}{u_A} = 35$$

$$\frac{60\ell + 4\ell}{20} - \frac{61\ell}{30} = 35 \Rightarrow \ell = \frac{1050}{35}$$

$$\Rightarrow L = 60\ell = \frac{1050}{35} \times 60 = 1800 \text{ m}$$

87. (d) $\frac{v_B}{v_A} \rightarrow \frac{v_A}{v_A}$

$$\text{Velocity of train A, } V_A = 90 \text{ km/h} = -2.5 \text{ m/s}$$

$$\therefore 1 \text{ km/h} = \frac{5}{18} \text{ m/s}$$

Velocity of train B, $V_B = 54 \text{ km/h} = 15 \text{ m/s}$

Velocity of train B w.r.t. train A

$$\vec{V} = \vec{V}_B - \vec{V}_A = 15 - (-25) = 40 \text{ m/s}$$

Time of crossing

$$(t) = \frac{\text{length of train}(l)}{\text{relative velocity}(V)}$$

$$(8) = \frac{\ell}{40} \Rightarrow \ell = 8 \times 40 = 320 \text{ m}$$

88. (d) $X_p(t) = at + \beta t^2$

$$V_p(t) = \frac{dX_p(t)}{dt} = a + 2\beta t \quad \dots (i)$$

$$X_Q(t) = ft - f^2$$

$$V_Q(t) = \frac{dX_Q(t)}{dt} = f - 2t \quad \dots (ii)$$

According to the question (i) = (ii)

$$a + 2\beta t = f - 2t \Rightarrow t(2\beta + 2) = f - a$$

$$\therefore t = \frac{f-a}{2(\beta+1)}$$

89. (d) $V_A = 36 \text{ km/hr} = 10 \text{ m/s}$

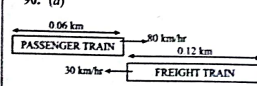
$$V_B = -72 \text{ km/hr} = -20 \text{ m/s}$$

$$V_{AB} = -1.8 \text{ km/hr} = -0.5 \text{ m/s}$$

$$V_{\text{man}} = V_{\text{man,A}} + V_{A,B}$$

$$= -0.5 + 10 - (-20) = 29.5 \text{ m/s}$$

90. (a)



Time taken if both moving in same direction

$$t_1 = \frac{\text{distance}}{\text{speed}} = \frac{0.12 + 0.06}{80 - 30} = \frac{0.18}{50}$$

Time taken if moving in opposite direction.

$$t_2 = \frac{\text{distance}}{\text{speed}} = \frac{0.12 + 0.06}{80 + 30} = \frac{0.18}{110}$$

$$\frac{t_1}{t_2} = \frac{11}{5}$$